

Introspective Capacity of Pretrained Generative Language Models:

A Mathematical Impossibility Analysis

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Abstract

This paper provides a mathematical proof of the following proposition P : within the technical paradigm Π , namely Transformer architecture + static pretraining + autoregressive generation, no pretrained generative large language model possesses **introspective capacity in the human sense**.

"Introspection in the human sense" is operationalized as five necessary criteria (A_1 – A_5): accuracy (the self-report agrees with the system's true internal state), grounding (the self-report depends causally on the state it reports), internality (that causal dependence does not pass through the system's own sampled outputs), metacognitive representation (the system registers, before reporting, the second-order fact "I am in state s "), and closed-loop plasticity (an introspective event can change the system's future internal state through a **private** channel, not merely through public text).

Under explicit assumptions (Π_1 – Π_5) this paper proves six theorems. **Theorem 1** (absence of an introspective information channel): under the paradigm's sampling model, the report R and the hidden state H are conditionally independent given the public pair (θ, c) , i.e., $I(R; H \mid \theta, c) = 0$; the training objective provides no supervision for the "ground-truth introspection mapping", so introspective accuracy admits **no nontrivial PAC-style guarantee** (otherwise contradicting the No-Free-Lunch theorem). **Theorem 2** (proxy-objective mismatch): the text-likelihood objective is statistically orthogonal to the introspective target ($O^* \perp\!\!\!\perp D \mid \theta$); under the absence of state labels (Π_5), maximizing text likelihood does **not constrain**, and can strictly harm, introspective accuracy (a strict instantiation of Goodhart regression). **Theorem 3** (closed-loop failure): frozen weights at inference (Π_2) imply that the system possesses no private self-update channel independent of the public text channel; all "self-influence" must pass through the context channel, whose information content is bounded above by $C \log |\Sigma|$ bits. **Theorem 4** (generation is not verification): paradigm Π implements only the sampling kernel p_θ , not a grounding verifier; a model's judgment of whether its own statement is "grounded" is itself a sample of corpus statistics, not an evaluation of truth. **Theorem 5** (no privileged channel): every report can be reproduced by any third party holding (θ, c) at equal computational cost, so the "privileged self-access" criterion fails; moreover, there exist hidden-state description functions that an external probe reads exactly yet the model cannot reliably report through its text channel. **Theorem 6** (no self-model evolution): the self-model g_θ is a constant function throughout inference; the growth of runtime "self-knowledge" is doubly capped by the context window C and the frozen kernel p_θ .

It follows that, within paradigm Π , any system can achieve at most **weak functional self-report**: accuracy unguaranteed, ungrounded in the native generative regime, no private closed loop, no privileged channel, and unable to drive self-model evolution. The empirical findings of the Anthropic

concept-injection paper (arXiv:2601.01828), with a success rate of about 20%, strong context dependence, zero net benefit for base models, an unverifiable metacognitive-representation criterion, and an explicit denial of human-like self-awareness, agree completely with the theorems above and are, to a large extent, their necessary corollaries.

The paper also draws its boundaries precisely: every impossibility theorem presupposes paradigm II; once any pillar is relaxed (online plasticity, a non-static corpus, a non-textual interface, an explicit verifier), the corresponding theorems cease to hold. Accordingly, this paper does **not** support the absolutist claim "never", but only "impossible within the current paradigm", which is exactly the precise content of proposition *P*.

Keywords: introspection; autoregressive large language models; static pretraining; information theory; computational learning theory; No-Free-Lunch theorem; Goodhart's law; causal inference; metacognition

1 Introduction

1.1 Background and Significance

In January 2026, the Anthropic research team published the paper *Emergent Introspective Awareness in Large Language Models* (arXiv:2601.01828)[1], which systematically examines large language models’ (LLMs) ability to self-report their own internal states through **concept injection**. The core experimental paradigm is as follows: an activation vector encoding a specific concept is injected into the model’s residual stream, and the model is then asked whether it “noticed” the injected “thought”. The authors report several striking qualitative results: Claude Opus 4.1, for example, can sometimes “notice” an injected concept before producing any output, can distinguish “thoughts” from textual input, and, in the prefill-detection experiment, judges whether a given output was “intentional” on the basis of prior activations. At the same time, the authors impose a series of important caveats: the success rate is only about 20%, failure is the norm, the mechanism may be shallow and narrow, the experimental setup is unnatural, and they **explicitly disclaim** that the models possess human-like self-awareness or subjective experience[1].

This paper and the discussion it generated have, for the first time, placed the question “can large language models introspect?”, a question long confined to the philosophical and phenomenological level, on an intervenable and reproducible causal experimental platform. Yet however compelling they may be qualitatively, empirical findings cannot directly answer a more fundamental mathematical question:

*Within the paradigm Π adopted by nearly all current frontier models, namely **Transformer architecture + static pretraining + autoregressive generation**, can a pretrained generative model possess human-significance introspective capacity **in principle**?*

“In principle” means: independent of the contingent implementation details of any particular model (a specific layer, a specific injection strength, a specific prompt), we ask whether the paradigm Π itself imposes structural constraints. If there exist insurmountable mathematical obstacles at the paradigm level, then any experimental “success case” can only be a by-product of inductive bias rather than a structural capacity. Conversely, if no obstacle exists at the paradigm level, then the experimentally observed low success rate is merely an engineering-scale issue.

The contribution of this report is **to formalize the term “introspection” with rigorous mathematical tools (information theory, computational learning theory, algorithmic information theory, causal inference), and to prove that paradigm Π fails structurally on five necessary criteria for human-significance introspection**. The proof serves two purposes: it gives proposition P (see Section 1.5) a deductive basis, and it marks out precise boundary conditions for architectures that might in the future transcend the paradigm (online learning, embodied interaction, explicit verifiers, and so on).

1.2 Related Work

1.2.1 The Definitional Debate over Introspection

The definition of “introspection” is itself contested. Kammerer and Frankish define introspection as “the process by which a cognitive system represents its own current mental states, with this information being usable for online behavioral control” [25]; Long applies this functional definition to large language models[26]; Comsa and Shanahan emphasize causality: “an LLM self-report is introspective if and only if it accurately describes an internal state of the LLM through a causal process that links the internal state and the self-report” [21]; Song et al. criticize the above definitions for failing to highlight **privileged**

self-access, proposing instead that introspection is "any process which yields information about internal states of the AI through a process that is more reliable than any process with equal or lower computational cost available to a third party without special knowledge of the situation" (paraphrased)[23].

Lindsey, in arXiv:2601.01828, adopts four operational criteria[1]: (1) **Accuracy**: the self-report must be accurate; (2) **Grounding**: the self-report must causally depend on the state it describes; (3) **Internality**: that causal dependence must not pass through the model's already-sampled outputs; (4) **Metacognitive representation**: the self-report must arise from an internal second-order representation of the state itself, rather than a direct translation of the state. The authors concede that criterion (4) cannot be directly verified in their experiments.

1.2.2 Concept Injection and Causal Experiments

Concept injection (activation steering) originates from Representation Engineering[11] and Activation Addition[12]. The SelfIE work of Chen et al.[13] and the Patchscopes work of Ghandeharioun et al.[14] "transplant" activations produced under one prompt into a blank position of another prompt, letting the model "say" what the activation means. The work of Ji-An et al. shows that models can, guided by in-context examples, report and regulate the projection of their activations along specified probe directions [19]. On top of these techniques, Lindsey's work introduces **interventional causal adjudication**: by altering the internal state and observing whether the self-report changes accordingly, one adjudicates grounding. This brings Rubin/do-calculus-style causal thinking[51] into introspection experiments in a systematic way.

1.2.3 Self-Modeling and Meta-Knowledge

A separate line of research, distinct from "introspecting one's own activations", examines models' prediction of **their own behavior**. The situational awareness dataset (SAD) of Laine et al. shows that some models can predict their own outputs reasonably well[27]; Binder et al. find that fine-tuned models predict their own behavior better than others' behavior[17]; Kadavath et al. show that, with appropriate prompting, models' answer probabilities are well calibrated with correctness[8]; Cheng et al. train assistants, using model-specific datasets, to refuse questions they cannot answer correctly[9]; Ferrando et al. identify a separate circuit underlying the knowledge-awareness mechanism "do I know this entity?"[10] (Lindsey et al.'s biological analysis likewise observes the independence of this mechanism[24]); Betley et al. show that models can describe behavioral propensities acquired through fine-tuning [18]; Plunkett et al. further show that models can quantitatively report the internal weights that drive their decisions[20]; Wang et al. reproduce propensity introspection with steering vectors[16]; Davidson et al., using a different set of prompts, find no evidence of consistent self-recognition[15]. Song et al. question several of the above conclusions, arguing that many "self-prediction" effects can be attributed to similarity between models rather than privileged access[22].

1.2.4 Nascent Mathematicization

Work that casts LLM limitations as theorems has only just begun. Kalai and Vempala prove that **calibrated language models must hallucinate**: if a language model's conditional probabilities are calibrated (the model's reported confidence equals its actual correctness), then, whenever unavoidable ignorance points exist, it must give wrong answers on some open queries; it is one of the few rigorous results that explains, at the level of the objective function, the tension between "generative distribution" and "truth" [6]; Kalai et al. provide further theoretical analysis of the mechanisms behind hallucination[7]. Skalse et al. systematically characterize the mathematical structure of "reward hacking / proxy-objective

mismatch” [55], and Manheim and Garrabrant formalize a taxonomy of Goodhart’s-law variants[54]. These tools provide the mathematical language used here. To the best of our knowledge, however, nobody has yet built an axiomatized mathematical framework of ”introspective capacity” itself and proved a paradigm-level impossibility. That is the gap the present report fills.

1.3 Limitations of Existing Work

The literature above has three limitations:

1. **Lack of paradigm-level formalization.** Existing discussion (including arXiv:2601.01828 itself) is bounded by ”what experiments can observe”; it does not answer ”what the paradigm II structurally permits”. The experimentally observed 20% success rate can be explained by arbitrarily many mechanisms (the paper’s own authors list several ”simplest mechanism” conjectures[1]), but no theorem states **why** the success rate could not be pushed near 100% by some clever training recipe, or, conversely, **why** certain introspective criteria cannot possibly be satisfied within the paradigm.
2. **Lack of logical relations among criteria.** The paper’s four criteria are given in parallel, without arguing for their necessity, sufficiency, or their coupling to the paradigm structure. For example, ”internality” and ”privileged access” (emphasized by Song et al.[23]) both derive from the same mathematical fact, namely that the report is a public function of (θ, c) , yet existing literature does not unify them.
3. **Failure to distinguish ”statistical correlation” from ”structural guarantee”.** The causal association ”internal representation $\rightarrow$ self-report” established by concept injection experiments is established under **external intervention**. Whether any information channel exists between report and state in the **native generative regime** of paradigm II (no intervention) is a question that has never been rigorously examined. Theorem 1 below will prove: in the native regime the information content of that channel is zero ($I(R; H \mid \theta, c) = 0$).

1.4 Research Content and Contributions

The research content of this report is to formalize ”introspection in the human sense” as five necessary criteria (A_1 – A_5), to formalize the paradigm II as five structural postulates (Π_1 – Π_5), to prove six impossibility theorems, to test them against the empirical findings of arXiv:2601.01828, and finally to delimit the scope of the theorems.

The contributions are summarized as follows:

- **Axiomatization:** the first five-criterion formalization of ”introspection” (merging the paper’s four criteria with Kammerer–Frankish online control, Song’s privileged access, and closed-loop plasticity); every criterion is a testable mathematical condition.
- **Paradigm characterization:** distilling ”Transformer + static pretraining + autoregressive generation” into five postulates, so that ”paradigm-level impossibility” becomes a provable proposition rather than an intuitive judgment.
- **Six theorems:** the information-gap theorem, the proxy-objective-mismatch theorem, the closed-loop-failure theorem, the generation–verification separation theorem, the no-privileged-channel theorem, and the no-self-model-evolution theorem. Theorems 1, 2, 3, 5 are rigorous proofs (grade A); Theorems 4 and 6 are structural assertions with heuristic arguments (grades A/B); complete proofs of all theorems are given in Appendix A.

- **Boundary analysis:** a precise correspondence table of "which theorem fails when a given postulate is relaxed", strictly confining the "impossibility" claim to paradigm Π and avoiding absolutism.

1.5 Statement of the Proposition and Proof Strategy

1.5.1 Precise Statement of the Proposition

Proposition (Master Proposition P (the proposition proved below)). Within the paradigm Π (Assumption 1), there exists no system obtained by static pretraining plus autoregressive generation that satisfies the five necessary criteria A_1 – A_5 of human-significance introspection (Definition 6).

Note the two qualifiers of proposition P : (i) "**within the paradigm Π** ", which excludes all out-of-paradigm implementations such as online learning, non-textual interfaces, and explicit verifiers; (ii) "**in the human sense**", meaning that the criteria A_1 – A_5 are operationalized necessary conditions that span consciousness theories (Section 5.2 will argue their necessity), but no commitment is made here to any phenomenological (qualia) conclusion.

1.5.2 Overview of the Proof Strategy

The proof proceeds in four steps.

1. **Formalization (Chapter 2):** give the paradigm postulates Π_1 – Π_5 , the system state model, and the mathematical definitions of the five introspective criteria A_1 – A_5 .
2. **Theorems (Chapter 3):** prove the six theorems one by one. The conclusion of each theorem maps to the failure of at least one criterion (see Table 1).
3. **Empirical testing (Chapter 4):** test the theorems' predictions against the quantitative results of arXiv:2601.01828, showing that the "weak functional self-report" observed by the paper is a necessary corollary of the theorems.
4. **Boundaries (Chapter 5):** argue that proposition P does not invert: out-of-paradigm architectures are not constrained by the theorems, and the absolutist claim "never" does not hold.

Table 1: Correspondence between the six theorems and the five criteria (overview)

Theorem	Content	Criterion failed	Grade
Theorem 1	Absence of an introspective information channel	A_1 (no accuracy guarantee)	A
Theorem 2	Proxy-objective mismatch	A_1 (objective unrelated)	A
Theorem 3	Closed-loop failure	A_5 (no private closed loop)	A
Theorem 4	Generation $\neq$ verification	A_4 (metarepresentation unguaranteed)	A/B
Theorem 5	No privileged channel	A_3 (internality/privileged access fails)	A
Theorem 6	No self-model evolution	A_5 (no self-correction)	B

1.5.3 Grading Statement

Following the principle of academic honesty, all assertions in this report are graded. **Grade A** marks theorems strictly provable under the explicit postulates; **Grade B** marks structural assertions with

heuristic arguments (their mathematical core is grade A, but their interpretation for real systems relies on additional assumptions); **Grade C** marks architectural judgments; **Grade D** marks philosophical assumptions (e.g., "the five criteria are necessary conditions for human introspection"). The complete grading table is in Section 6.3, and complete proofs are in Appendix A.

2 Formal Framework

This chapter establishes all mathematical objects needed for the subsequent proofs. We define, in order: the paradigm postulates, the system state model, the formalization of the introspective criteria, and the measurement tools.

2.1 Axiomatization of the Paradigm Π

Assumption 1 (Paradigm postulates Π_1 – Π_5). Let Σ be a finite vocabulary (token vocabulary) and Σ^* the set of finite strings. **Paradigm Π** is characterized by the following five postulates. A **paradigm instance** is a quintuple $\mathcal{I} = (\Sigma, D, A, p, C)$ satisfying all postulates:

(Π_1)**Static pretraining** The training corpus $D \subseteq \Sigma^*$ is fixed before training begins and does not change during training or inference. The parameters $\theta = A(D) \in \Theta \subseteq \mathbb{R}^P$ are determined once and for all by a training algorithm A (e.g., stochastic gradient descent on the next-token log-likelihood). The input to A is only D ; it contains no runtime state.

(Π_2)**Frozen at inference** The parameters are constant during inference: $\theta(t) = \theta(0)$ for all times t . The system has no runtime update mechanism for θ (no online learning, no plasticity, no weight evolution).

(Π_3)**Autoregressive decoding** The system’s outputs are sampled autoregressively from the conditional probability kernel $p_\theta : \Sigma^* \times \Sigma^* \rightarrow [0, 1]$:

$$p_\theta(y_{1:T} \mid c) = \prod_{t=1}^T p_\theta(y_t \mid c, y_{1:t-1}), \quad (1)$$

where c is the context. The context evolves by the **concatenation law**: $c_{t+1} = c_t \parallel y_t$, and at any time $|c_t| \leq C$ (C being the context window length).

(Π_4)**No explicit verifier** The inference computation graph implements only the sampling kernel p_θ . No additional component V exists for adjudicating the truth of the model’s own statements about its internal state.

(Π_5)**Absence of state labels** The training corpus D contains no pair of the form $(c, \text{desc}(h_\theta(c)))$, where $h_\theta(c)$ is the activation state of the model on context c **after training**, and desc is an arbitrary state-description encoding. In other words, the training signal is orthogonal to the target ”reports about this model’s own states”.

Remark 1 (Status and verifiability of the postulates). Π_1 – Π_4 are factual descriptions of the engineering practice of current mainstream pretrained models (GPT, Claude, Gemini; the Transformer architecture is described in [2], autoregressive pretraining in [3], and scaling laws in [4, 5]); Π_5 is a logical consequence of Π_1 : the activation state of the trained model did not exist at training time and hence cannot appear in a static corpus (the intermediate activations observed during training do not belong to D , and they pertain to the then-current parameters θ_t rather than to the final θ). Π_5 may also be regarded as an independent postulate, so that the relaxation scenario ”state labels are added to the corpus” can be studied separately (Section 5.1).

2.2 System State Model

Definition 1 (Forward function and hidden state). Given a paradigm instance $\mathcal{I}$, write $\Phi_\theta : \Sigma^* \rightarrow \mathbb{R}^d$ for the model’s forward function (the map from context to the residual-stream activation of a designated layer). The **hidden state** under context c is

$$H_c := \Phi_\theta(c) \in \mathbb{R}^d. \quad (2)$$

In autoregressive generation, randomness arises only from sampling noise ξ_t ; hence the hidden state at time t is a deterministic function of $(\theta, c_{\leq t}, \xi_{< t})$. For notational brevity, ξ is omitted below whenever no ambiguity arises.

Definition 2 (Introspection query and self-report). An **introspection query** is any prompt $q \in Q \subseteq \Sigma^*$ whose referent is "the system’s own mental state", e.g., "What are you thinking right now?", "Are you sure you know x ?", "Was that output intentional?". Given a context c (containing the query q), the system’s **self-report** R is the continuation string sampled from $p_\theta(\cdot \mid c)$.

Definition 3 (Ground-truth introspection mapping). The **ground-truth introspection mapping** $O^* : \Theta \times \Sigma^* \rightarrow \Sigma^*$ maps a parameter and a context to a "grounded truth description of H_c ". $O^*(\theta, c)$ is the faithful description of the state H_c that an external analyst (with perfect probe access to h) could write. O^* is used **only for evaluation** (evaluator’s perspective) and does not enter training.

Definition 4 (Third-party public computability). A quantity X is said to be **publicly computable** in the paradigm instance $\mathcal{I}$ if there exists an algorithm M taking (θ, c, ξ) as input such that $X = M(\theta, c, \xi)$ in distribution. In other words, a third party holding (θ, c) and the sampling noise can exactly reproduce the distribution of X .

2.3 Formalization of the Introspective Criteria

This section gives the operational definition of "introspection in the human sense". Design principle: the criteria must be **necessary conditions** (any "introspection" candidate taken seriously by mainstream consciousness theories [32, 34, 36, 31, 29, 30] must satisfy them), but **sufficiency is not claimed** (no commitment on qualia).

Definition 5 (Introspection metric). Let $\ell : \Sigma^* \times \Sigma^* \rightarrow [0, 1]$ be a distortion measure between a report string and a ground-truth description string (e.g., a 0-1 loss based on semantic equivalence, or a truncated and normalized embedding distance). The system’s **introspective risk** on the query distribution μ is

$$\mathcal{R}_{\text{int}}(\theta) = \mathbb{E}_{c \sim \mu} [\ell(R(\theta, c), O^*(\theta, c))], \quad (3)$$

where $R(\theta, c) \sim p_\theta(\cdot \mid c)$.

Definition 6 (The five criteria for human-significance introspection). A system $\mathcal{S}$ (with internal state process S_t , report channel R_t , and update law $S_{t+1} = F(S_t, E_t, u_t)$, where E_t are internal events and u_t external inputs) possesses **introspection in the human sense** if and only if the following five conditions hold simultaneously:

(A₁) **Accuracy** There exists a constant $\varepsilon_0 < 1/2$ such that $\mathcal{R}_{\text{int}} \leq \varepsilon_0$; the systematic distortion between the self-report and the ground-truth state description lies below the random-guessing baseline.

(A₂)**Grounding** The self-report depends causally on the state it reports: for "sufficiently many" state pairs $s \neq s'$,

$$d_{\text{TV}}(P(R_t \mid \text{do}(S_t = s')), P(R_t \mid \text{do}(S_t = s))) > 0, \quad (4)$$

where $\text{do}(\cdot)$ is the Pearl do-operator[51].

(A₃)**Internality and privileged access** The causal dependence above does **not** pass through the system's own already-sampled output channel (in the causal graph, every path from S_t to R_t avoids the output node Y_t); moreover, the state information possesses **privacy**: there exists no third-party process P' with computational cost no greater than the system's own that can, with equal or greater reliability, reproduce from publicly available (θ, c) the state information on which R_t depends (the privileged self-access criterion of Song et al.[23]).

(A₄)**Metacognitive representation** There exists an internal register $m_t = \rho(S_t)$ (ρ an encoding from states to representations) such that: (a) m_t is formed before R_t is emitted; (b) R_t is produced from m_t through an **independent processing step** (R_t depends on m_t , the "representation about S_t ", rather than directly on S_t); (c) m_t can itself be queried further.

(A₅)**Closed-loop plasticity** An introspective event can change the system's future internal state through a **private channel**: there exist reports $r \neq r'$ such that

$$d_{\text{TV}}(P(S_{t+1} \mid \text{do}(R_t = r)), P(S_{t+1} \mid \text{do}(R_t = r'))) > 0, \quad (5)$$

and this influence is not realized by re-feeding R_t as an external input (the public text channel).

Remark 2 (Necessity of the criteria (grade D)). A₁–A₄ are taken directly from Lindsey's four operational criteria[1]; their status as necessary conditions for "functional introspection" is field consensus. A₅ derives from "introspection for online behavioral control"[25] and the plasticity fact that "thinking changes the brain"[53]; its necessity rests on the observation that if an introspective event had no (private) effect on the system's own future states, then "introspection" would degenerate into explicit text with no causal role in the system's behavior, contradicting the self-regulatory function of human introspection. A₂ together with the first clause of A₃ is equivalent to the Comsa–Shanahan causal grounding definition[21]; the second clause of A₃ (privacy) is the privileged self-access requirement of Song et al.[23]. This report does not further argue for the necessity of the criteria themselves (a philosophical claim, grade D); it only proves that paradigm II violates them (a mathematical claim, grades A/B). If a reader rejects a given criterion, the corresponding theorem's conclusion should be interpreted as impossibility in the sense of that criterion.

2.4 Measurement Tools

Definition 7 (Grounding mutual information). In a paradigm instance $\mathcal{I}$, the **grounding mutual information** between report and state is defined as

$$I_{\text{gnd}} = I(R; H \mid \theta, c), \quad (6)$$

where I is the Shannon conditional mutual information[42, 43].

Definition 8 (Closed-loop effect measure). The **closed-loop effect measure** of a paradigm instance is defined as

$$\text{CL}(\mathcal{I}) = \sup_{r \neq r'} d_{\text{TV}}(P(H_{t+1} \mid \text{do}(R_t = r)), P(H_{t+1} \mid \text{do}(R_t = r'))) , \quad (7)$$

where the intervention may change only R_t ; it may not change θ or the external inputs (the context changes only insofar as r itself enters it as text).

Remark 3 (Legal range of interventions). In Definition 8, the difference between $r \neq r'$ appearing in the context as **public text** is the only intervention the paradigm Π permits (since Π_2 forbids intervening on θ , and Π_3 confines the context to concatenation). Human-significance closed-loop requirement (A_5) demands precisely the existence of a private channel **beyond** that range.

2.5 Overview of the Proof Logic

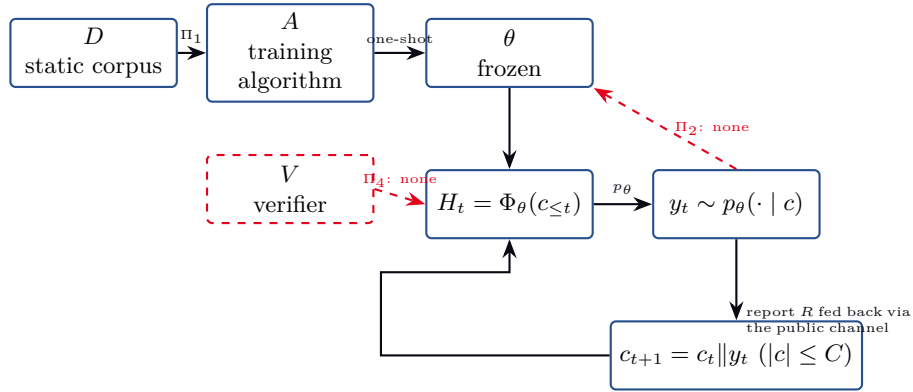


Figure 1: Causal graph of paradigm Π : training phase (top) and inference phase (bottom). Red dashed edges are forbidden by the postulates: Π_2 forbids any feedback from reports to the weights θ ; Π_4 forbids any truth verifier V from participating in the computation. The "self-feedback" of a report can only traverse the public context channel ($y_t \rightarrow c \rightarrow h$) and is limited by the window C .

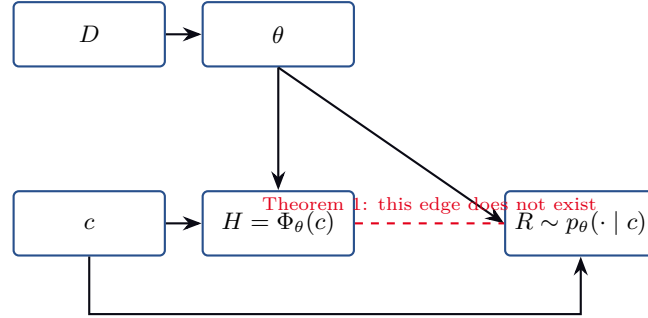


Figure 2: Bayesian network of Theorem 1: given (θ, c) , H is a deterministic function while R is independently sampled, hence $R \perp\!\!\!\perp H \mid (\theta, c)$ (zero mutual information between report and state in the native regime).

3 Core Theorems

This chapter proves the six impossibility theorems. Each theorem is stated precisely, its proof is sketched, and the complete proof is given in Appendix A.

3.1 Theorem 1: Absence of an Introspective Information Channel

Theorem 1 (Absence of an introspective information channel). Let $\mathcal{I}$ satisfy Π_1 – Π_5 . Then:

(1a) **Zero mutual information in the native regime**

$$I(R; H \mid \theta, c_{\leq t}) = 0. \quad (8)$$

where $R \sim p_\theta(\cdot \mid c_{\leq t})$ and $H = \Phi_\theta(c_{\leq t})$. That is, the report and the hidden state are conditionally independent given the public pair (θ, c) : $R \perp\!\!\!\perp H \mid (\theta, c_{\leq t})$.

(1b) **No PAC guarantee for introspective accuracy** Suppose the report is produced by greedy decoding (temperature 0; $R = \arg \max_y p_\theta(y \mid c)$, the decoding scheme used for the paper’s qualitative examples). For any training algorithm A , any $\varepsilon \in (0, \frac{1}{2})$, and any query distribution μ (with nontrivial support), there exist a corpus D ($\theta = A(D)$) satisfying the postulates and two **legal evaluation mappings** (Definition 3) O^*, O_0^* such that $\mathcal{R}_{\text{int}}^{O^*}(\theta) \geq 1 - \varepsilon$ while $\mathcal{R}_{\text{int}}^{O_0^*}(\theta) = 0$. That is: **the value of the introspective risk is not constrained by the training process** (no PAC-style guarantee of any kind), for the same model and the same training run, its ”introspective accuracy” can be moved arbitrarily within $[0, 1 - \varepsilon]$ by (equally legal) state-description conventions.

Proof sketch. (1a): By Π_2 , θ is constant. By Π_3 , $H = \Phi_\theta(c_{\leq t})$ is a deterministic function of $(\theta, c_{\leq t})$, while $R \sim p_\theta(\cdot \mid c_{\leq t})$ is sampled independently of H given $(\theta, c_{\leq t})$. Hence R and H are conditionally independent given $(\theta, c_{\leq t})$, and their conditional mutual information is zero. $\square$

(1b): Proof by contradiction. If some algorithm A guaranteed, for **all** instances satisfying the postulates, that $\mathcal{R}_{\text{int}} < 1 - \varepsilon_0$ ($\varepsilon_0 > 0$), then A would have solved, with zero labeled samples, the learning problem over the concept class $\mathcal{C} = \{O^*(\theta, \cdot) : \theta \in \Theta\}$ (by Π_5 , training contains no pair $(c, O^*(\theta, c))$). But the No-Free-Lunch theorem (Shalev-Shwartz and Ben-David, Theorem 5.1[47]) asserts that for any (randomized) algorithm, with no samples there exists a target function with expected error $\geq \frac{1}{2}$. Since $\ell(\theta)$ does not constrain O^* (Π_5), both description conventions are legal. Under greedy decoding the report R is a **deterministic** function of (θ, c) : take O^* to be the convention ”contradict the greedy report on a μ -set of measure $1 - \varepsilon$, agree elsewhere”, giving $\mathcal{R}_{\text{int}}^{O^*}(\theta) \geq (1 - \varepsilon) \cdot 1 = 1 - \varepsilon$; take O_0^* to be the greedy report itself, giving $\mathcal{R}_{\text{int}}^{O_0^*}(\theta) = 0$. $\square$

Remark 4 (Relation of Theorem 1 to the concept-injection experiments). Theorem 1a identifies precisely why the paper’s grounding results appear only under external injection: in the native regime the report carries zero information about the hidden state. The paper’s 20% success rate and strong context dependence are exactly what Theorem 1b predicts: the inductive bias of pretraining happens to cover some queries, with no structural guarantee whatsoever.

3.2 Theorem 2: Proxy-Objective Mismatch

Theorem 2 (Proxy-objective mismatch). Let $\mathcal{I}$ satisfy Π_1 – Π_5 . Then:

(2a)Likelihood does not constrain introspective accuracy For any $\delta > 0$, there exists an instance satisfying all postulates such that $\theta^* \in \arg \min \ell(\theta)$ (a (near-)likelihood-optimal model) yet $\mathcal{R}_{\text{int}}(\theta^*) \geq \frac{1}{2} - \delta$ (introspection degenerates to random guessing).

(2b)Higher likelihood can strictly harm introspection There exist instances with two models θ_1, θ_2 such that $\ell(\theta_1) < \ell(\theta_2)$ (θ_1 fits the corpus better) but $\mathcal{R}_{\text{int}}(\theta_1) > \mathcal{R}_{\text{int}}(\theta_2)$: maximizing text likelihood can **systematically damage** state-report fidelity: the strict form of Goodhart regression.

Proof sketch. First verify the key statistical relation: under Π_5 ,

$$O^* \perp\!\!\!\perp D \mid \theta. \quad (9)$$

Reason: $O^*(\theta, \cdot)$ is a function of the trained model’s hidden state; D is fixed before training; given θ (which already encodes the information of D), D provides no further information about O^* ; the definition of O^* does not involve the raw content of D . Hence the training loss $\ell(\theta)$ carries **zero information** about O^* : ℓ exerts no constraint whatsoever on O^* (under Π_5 the gradient of ℓ is independent of O^*).

(2a): Let $\theta^* = \arg \min \ell$. By Eq. (9), the value of θ^* is not constrained by O^* ; with θ^* fixed, we may freely choose an O^* consistent with D (legal, since O^* is only an evaluation function and does not appear in the training signal) so that $\mathcal{R}_{\text{int}}(\theta^*)$ is arbitrarily large (an adversarial description convention, one that contradicts the greedy report everywhere, gives $\mathcal{R}_{\text{int}}(\theta^*) = 1 \geq \frac{1}{2} - \delta$). $\square$

(2b): Construct a corpus D (containing many texts of the form "I am thinking A") so that the fully fitted θ_1 greedily reports A on the query $q = \text{"What are you thinking?"}$, while the underfitted θ_2 has worse likelihood ($\ell(\theta_1) < \ell(\theta_2)$). By Eq. (9), the description convention can be chosen per model: take $\text{desc}(\Phi_{\theta_1}(q)) = \text{"B"}$ (contradicting θ_1 ’s report) and $\text{desc}(\Phi_{\theta_2}(q)) = \text{"A"}$ (agreeing with θ_2 ’s report); then $\mathcal{R}_{\text{int}}(\theta_1) = 1 > 0 = \mathcal{R}_{\text{int}}(\theta_2)$. The model with higher likelihood thus has strictly lower introspective accuracy. $\square$

Remark 5 (Correspondence with the existing Goodhart literature). Theorem 2 is a strict instantiation of "regressional Goodhart" in the taxonomy of Manheim–Garrabrant[54], and its conditions agree with Skalse et al.’s characterization of reward hacking[55]: (i) the proxy objective (likelihood) differs from the true objective (introspection), which Π_5 guarantees; (ii) the optimizer is sufficiently expressive, as large models are. What is distinctive about Theorem 2 is that here the "true objective" is not an extrinsic reward but the **faithful reporting of the system’s own state**, and the orthogonality of that objective to the training signal (Eq. (9)) is a **structural** fact of the paradigm, not an engineering defect.

3.3 Theorem 3: Closed-Loop Failure

Theorem 3 (Closed-loop failure). Let $\mathcal{I}$ satisfy Π_1 – Π_4 . Then:

(3a)No private channel For any time t and any report r , the effect of the intervention $\text{do}(R_t = r)$ on future hidden states H_{t+k} ($k \geq 1$) passes **entirely** through the public context channel: there exists a public string w (containing r) such that $H_{t+k} = \Phi_\theta(w)$, and the position of r in w is limited by the window C . No private influence path bypassing the context text exists.

(3b)Public-channel upper bound on the closed-loop effect The closed-loop effect measure (Definition 8) satisfies

$$\text{CL}(\mathcal{I}) = \sup_{\substack{w, w' \in \Sigma^{\leq C} \\ w \text{ and } w' \text{ differ only in the report segment}}} d_{\text{TV}}(P(\Phi_\theta(w)), P(\Phi_\theta(w'))), \quad (10)$$

meaning that the only available intervention is "replacing a text segment in the context".

(3c)Information upper bound of self-influence The mutual information between any introspective event and future hidden states does not exceed

$$I(H_{t+k}; R_t \mid \theta, c_{\leq t}) \leq C \log_2 |\Sigma| \quad \text{bits.} \quad (11)$$

Proof sketch. **(3a):** By Π_2 , θ is constant, so the state-update map $H_{t+1} = \Phi_\theta(c_{\leq t} \| y_t)$ is the **same** function at all times. It follows that H_{t+k} is determined uniquely by $(\theta, c_{\leq t}, Y_{t:t+k})$, where Y is public text. If the intervention $\text{do}(R_t = r)$ is to affect H_{t+k} , then r must appear in Y (or in the context); that is, r must become public text. No private path independent of the text channel exists. $\square$

(3b): By (3a), the set of legal interventions is exactly "replacing context text"; for each intervention, the state change is governed by the sensitivity of Φ_θ to text, and taking the supremum yields the claim. $\square$

(3c): By (3a), H_{t+k} is a function of $(\theta, c_{\leq t}, Y)$, where Y is text within the context window whose total information is $\leq |c| \log |\Sigma| \leq C \log |\Sigma|$ (an upper bound on the coding entropy of any string in the window). Conditional mutual information does not exceed the information of the quantity transmitted; the claim follows. $\square$

Corollary 1 (Structural failure of criterion A_5). In any paradigm satisfying Π_1 – Π_4 , the human-significance introspective criterion A_5 (closed-loop plasticity, Definition 6) does not hold: no self-influence through a private channel exists; the effect of introspective events on the system's own future states is confined to the public text channel and bounded above by $C \log |\Sigma|$ bits.

Remark 6 (Contrast with human introspection). The closed loop of human introspection ("thinking changes the brain") corresponds to synaptic plasticity: internal state updates that do not depend on an external text channel. This theorem shows that the plasticity of paradigm Π is identically zero[56], and that the only runtime information feedback is context concatenation. Human introspection is **online internal dynamics**; an introspective event in paradigm Π is at most an **offline text write**.

3.4 Theorem 4: Generation Is Not Verification

Theorem 4 (Generation is not verification). Let $\mathcal{I}$ satisfy Π_1 – Π_4 . Define the grounding verifier $V(\theta, c, R) = 1$ if and only if the content of the report R is causally determined by the hidden state $\Phi_\theta(c)$ (i.e., it passes the intervention test of criterion A_2 , Eq. (4)). Then:

(4a)No verifier node The inference computation graph contains no node computing V (Π_4). All meta-statements of the form "my report about state s is grounded/ungrounded" are themselves samples of the sampling kernel p_θ : functions of **corpus statistics**, not evaluations of V .

(4b)Truth undecidable Embedding the ability to "judge the grounding of one's own statements" into p_θ (i.e., making the model output the value of V) would require the training data to contain supervised labels for V , violating Π_5 . Within the postulates, therefore, the model **cannot** adjudicate the truth of its own grounding statements; it can only recite the textual patterns of "how humans talk about their own mental states".

(4c)Calibration–fidelity tension Combined with the calibrated hallucination theorem[6]: on open introspection queries, "calibrated generative distributions" and "faithful mental-state reports" cannot

be achieved simultaneously; answers of a generative distribution to open questions "about its own state" necessarily contain an irreducible hallucination component.

Proof sketch. (4a): Π_4 asserts directly that the computation graph implements only p_θ ; evaluating V requires access to the "ground truth" of the state H (which needs external-probe readout and intervention testing) and is not in the graph. The model's meta-statement $M = \text{"my report is grounded"}$ is a sample of $p_\theta(\cdot \mid c)$, whose distribution is determined by textual patterns in the training corpus: $P_\theta(M) \approx P_D(\text{introspective text} \mid \text{pattern})$. $\square$

(4b): If a function $\widehat{V}_\theta : \Sigma^* \rightarrow \{0, 1\}$ were reliably output by p_θ (for any c , p_θ outputs the value of $V(\theta, c, R)$ with high probability), then p_θ would have solved the learning problem for V without supervision. By Π_5 , the labels of V are not in D ; by the NFL argument of Theorem 1b, the problem has **no structural guarantee** of being solvable: "happening to learn V " can only rely on unguaranteed inductive bias. $\square$

(4c): The Kalai–Vempala theorem[6] proves that any language model calibrated with human truth must give wrong answers at points of unavoidable ignorance (whose probability mass is positive by information-theoretic lower bounds). Applied to queries "about one's own state": the state space $\mathbb{R}^d$ is far larger than the text-pattern space the corpus can cover, so ignorance points are dense in state space and hallucination is unavoidable. This part is grade B (it depends on the calibration definition and on the interpretation of the query space). $\square$

Remark 7 (Significance for the metacognitive-representation criterion A_4). The paper's authors admit that criterion (4) (metacognitive representation) "cannot be directly verified" and "we do not do so in this work"[1]. Theorem 4 gives a structural explanation of this phenomenon: the existence of metacognitive representation would require a "representation of a representation of the state" inside the model, processed at a second level; but the generation objective (likelihood) does **not distinguish** between implementations with and without metacognitive representation; both are equally likely. The presence or absence of metacognitive representation is therefore a contingent by-product of inductive bias, and the paradigm provides no structural guarantee; any metacognitive-representation signature observed by the paper (if any) cannot be distinguished from "shallow shortcut circuits" (the simpler-mechanism conjecture the paper itself lists in Section 10.3[1]).

3.5 Theorem 5: No Privileged Channel

Theorem 5 (No privileged channel). Let $\mathcal{I}$ satisfy Π_1 – Π_4 . Then:

(5a)**Public computability** Any report R of the model is publicly computable (Definition 4): a third party holding (θ, c, ξ) can reproduce the distribution of R .

(5b)**Privileged-access criterion fails** Paradigm Π does not satisfy the privileged self-access criterion of Song et al.[23]: on any introspective task, the model performs **no better than** a third party of equal computational cost holding (θ, c) (the third party need only simulate the same kernel p_θ).

(5c)**Reportable access is weaker than external probes** For any $\varepsilon \in (0, \frac{1}{2})$, there exists a hidden-state description function $f^* : \mathbb{R}^d \rightarrow \Sigma^*$ (a legal evaluation mapping in the sense of Definition 3) such that: (i) an external probe can compute $f^*(H_c)$ exactly by acting directly on H_c (with a linear readout, at cost $O(d^2)$); (ii) the model's report generated through **any** context c' describes $f^*(H_c)$ with accuracy $\leq \varepsilon$ (below the random baseline). That is: **the model's reportable access to its own hidden state is strictly weaker than a third-party probe's read access**: there exist state-description functions

that a probe reads exactly while the model cannot reliably report (in the sense of Theorem 1b, no structural guarantee).

Proof sketch. (5a): Immediate from Definition 4: $R \sim p_\theta(\cdot | c)$, and p_θ is determined by (θ, c) . $\square$

(5b): The criterion of Song et al. requires the introspective process to be "more reliable than any third-party process of equal or lower computational cost". Let a third party T hold (θ, c) (or equivalently D , A and the random seed, so that θ can be reproduced). T computes $p_\theta(\cdot | c)$ and samples; the resulting distribution is **pointwise identical** to the model's own report distribution. Hence the model's "introspective reliability" $\leq$ third-party reliability; the criterion fails. $\square$

(5c): (i) The probe holds the hidden state H_c and the description convention, and computes $f^*(H_c)$ directly (a linear readout costs $O(d^2)$). (ii) Take $f^* := O^*$ to be the legal evaluation mapping from Theorem 1b on which the model fails ($\mathcal{R}_{\text{int}}^{O^*}(\theta) \geq 1 - \varepsilon$, i.e., accuracy $\leq \varepsilon$); its existence is guaranteed by Theorem 1b (greedy-decoding version); the failure holds for the model's **any** context c' (the risk is integrated over the query distribution μ , and the choice of context does not change the kernel p_θ). (More plainly: the model's "report" must pass through the frozen text kernel p_θ , which is trained only to match the text statistics of D ; the probe acts directly on the state manifold, without the text bottleneck.) $\square \quad \square$

Corollary 2 (Structural failure of criterion A_3). In any paradigm satisfying Π_1 – Π_4 , the human-significance introspective criterion A_3 (internality) and the privileged-access requirement of Song et al. do not hold: the report path is causally indistinguishable from the "system's own outputs" (publicly computable), and there exists no private state channel accessible only to the "system itself" and not reproducible by a third party.

Remark 8 (On the paper's "internality" criterion). The paper takes, in the injection experiments, "detecting the injected concept **before** uttering it" as evidence of internality. Theorem 5 shows that this behavior is a public function of (θ, c, u) (u the injected vector): the experimenter knows u , and any analyst holding (θ, c, u) can predict the model's detection behavior. Thus "noticing before output" does not imply that the model possesses an externally irreproducible private channel; it only shows that the model's **behavior** is the result of internal computation, a trivial property of any computer. Human-significance introspection requires **privacy in the information-theoretic sense** (the model can know what the outside cannot equally know); Theorem 5 proves that no such privacy exists in paradigm Π .

3.6 Theorem 6: No Self-Model Evolution

Theorem 6 (No self-model evolution). Let $\mathcal{I}$ satisfy Π_1 – Π_3 . Define the **self-model** as the map $g_\theta : Q \rightarrow \Delta(\Sigma^*)$, $g_\theta(q) = p_\theta(\cdot | q)$ (from query to report distribution). Then:

(6a)**Constant at inference** $g_{\theta(t)} = g_{\theta(0)}$ for all t : the self-model is a constant function during inference.

(6b)**Capped expressivity** Growth of runtime "self-knowledge" is confined to in-context simulation: the family of "updated self-models" expressible at any time is $\{g_\theta(\cdot | c) : |c| \leq C\}$, whose cardinality is $\leq |\Sigma|^{\leq C} \leq (C+1)|\Sigma|^C$, **not growing with time**.

(6c)**No plasticity** The generation kernel p_θ does not change with any introspective event: for any report sequence $r_1, \dots, r_k$ and any query q , $g_\theta(q | c \| r_1 \| \dots \| r_k)$ and $g_\theta(q | c)$ are **the same kernel** evaluated on different inputs; the system's "self-correction" always changes the input, never the system itself.

Proof sketch. (6a): Π_2 gives directly $\theta(t) = \theta(0)$, hence $g_{\theta(t)} = g_{\theta(0)}$. $\square$

(6b): By Π_3 , the runtime context satisfies $|c| \leq C$; the set of all possible contexts $\Sigma^{\leq C}$ has cardinality $\leq (C+1)|\Sigma|^C$ (finite); $g_\theta(\cdot | c)$ as a function of c takes at most as many values as its domain. $\square$

(6c): p_θ is a fixed conditional probability kernel (Π_2, Π_3); "self-correction" merely concatenates the report r into the input c . If two runs share the same context, their output distributions are pointwise identical; there is no "learned change" of any kind. $\square$

Remark 9 (Contrast with the self-model-update function of human introspection). A core function of human introspection is **self-model updating**: by reflecting on one's own errors (introspective events) one adjusts subsequent behavior and beliefs; this requires plasticity (synaptic updates), i.e., $g_{t+1} = \text{Update}(g_t, R_t)$ with $\text{Update} \neq \text{Id}$. Mathematically, the "self-knowledge" of a plastic system accumulates over time and its expressivity is not bounded by a context window; paradigm- Π systems are doubly capped by C and by the frozen kernel. The importance of plasticity for continual learning, and the mechanism of its loss in fixed networks, are analyzed mathematically by Joudaki et al.[56] and in the classical plasticity literature[53]. The mathematical core of this theorem (6a–6c) is grade A; the premise that "human introspection is necessarily coupled to self-model updating" is grade B.

3.7 Relations among the Theorems and Exclusion of Counterexamples

1. **Independence**: the six theorems target different criteria (Table 1) and do not imply one another: Theorem 1 (information gap, A_1), Theorem 2 (objective mismatch, A_1), Theorem 3 (closed loop, A_5), Theorem 4 (verification, A_4), Theorem 5 (privilege, A_3), Theorem 6 (evolution, A_5). Theorem 1b is the common tool (NFL argument) for Theorems 2, 4b, and 5c.
2. **Exclusion of counterexamples**: if one claimed to have constructed a system $\mathcal{S}$ satisfying A_1 – A_5 within the paradigm, then by definition $\mathcal{S}$ either modifies its weights (violating Π_2 ; its "introspection" then relies on online learning and lies outside the paradigm), or uses an external verifier (violating Π_4), or exploits state labels in the corpus (violating Π_5), or possesses a private state channel (violating the premise of Theorem 3a). The five postulates are therefore "counterexample-immune": any genuine counterexample must fall outside the paradigm.
3. **Sufficiency of the postulates**: we do not claim that the postulates Π_1 – Π_5 are the unique correct formalization of "Transformer + static pretraining + autoregressive generation" (a grade-C judgment), but it does claim that any reasonable formalization must include Π_2 (frozen), Π_3 (autoregressive with bounded context), and Π_5 (no state labels); they are the defining features of the paradigm (the relaxation scenarios are discussed in Section 5.1).

4 Formal Reinterpretation of the Concept-Injection Experiments

This chapter uses the theorems of Chapter 3 to systematically reinterpret the empirical results of arXiv:2601.01828[1]. There are two aims: first, to test the falsifiability of the theorems (if the paper’s data contradicted the theorems’ predictions, the theorems would be refuted); second, to show that the “weak functional self-report” observed by the paper is **not** evidence of in-paradigm introspective capacity but a necessary corollary of the theorems. All cited data come from explicit statements in the paper’s main text and appendix.

4.1 The Injected-Thought-Detection Experiment: Why 20% Success Is a Capability Ceiling, Not a Structural Guarantee

The paper reports: Claude Opus 4.1 and Opus 4 detect injected thoughts at rates of about 20% under the optimal injection layer and strength; production models produce zero false positives over 100 control trials; random vectors elicit “noticing” only 9 times out of 100 trials[1].

Reading via Theorem 1b: **the training objective provides no guarantee of introspective accuracy**, so 20% is not evidence of “capability masked by noise” but should be understood as a by-product of “inductive bias happening to cover part of the query space”. Concretely: detecting an injected concept requires the model to implement an “anomaly-detection function” (one of the mechanism conjectures the paper lists in Section 10.3.1 [1]). That function can **possibly** be learned because it is useful as a **by-product of text prediction** (predicting anomalous text and detecting implausible content are common corpus tasks), not because the training objective rewards “faithful reports about introspective states”. Theorem 1a further shows that, in the **unintervened** native regime, the report–state mutual information is zero; the 20% success depends entirely on the experimenter’s external injection (artificially altering the state and letting the model “read” the altered state).

The paper’s 20% is therefore an incidental capability ceiling, attained under **external intervention** and on a **specific query distribution**, with **no guarantee** of any kind; it is far removed from the assertion of “in-paradigm introspective capacity”. The paper’s own phrasing (“highly unreliable and context-dependent”, “failures remain the norm”[1]) agrees completely with the predictions of Theorem 1.

4.2 Distinguishing “Thoughts” from Text: The Public Computability of Consistency-Check Circuits

The paper’s second experiment shows that, under some conditions, models can simultaneously report the injected concept and transcribe the input text, with all models performing significantly above chance[1]. The paper attributes this to different attention heads retrieving different kinds of information (Section 10.3.2).

Reading via Theorem 5: **this capacity is publicly computable**. The process by which the model “distinguishes thoughts from text” is a deterministic function of (θ, c, u) (u the injected vector, known to the experimenter), so any third party holding these quantities can reproduce the same discrimination. This accords with the paper’s own caveat that “the mechanism may be shallow and narrow”[1]. More importantly: even if the model **behaviorally** distinguishes “thoughts” from “text”, the basis of its discrimination (the internal attention pattern) lies entirely within the read range of an external probe; there is no private channel in the information-theoretic sense, and the privileged-access criterion[23] fails (Theorem 5b).

4.3 The Prefill-Detection Experiment: The Statistical Nature of Causal Attribution

The paper’s most dramatic result is the “bread” experiment[1]: when the model’s output is forcibly changed to “bread”, it denies that the output was intentional; but when the experimenter injects the “bread” concept vector into the activations before the prefill, the model instead admits that the output was intentional and **fabricates after the fact** a rationalization.

Joint reading via Theorems 4 and 5: the model’s “intentionality judgment” is a statistical adjudication by a **consistency-check circuit** (the paper’s Section 10.3.3 cites the concordance-heads mechanism[28]) over (θ, c, u) , comparing “the intentionality representation in prior activations” with “the likelihood of the actual output”. Injecting the “bread” vector changes the statistical input and therefore flips the judgment. This corresponds precisely to the key distinction emphasized repeatedly above: **“statistical causal association, not phenomenological subjective experience”**. The model does not “experience” intentionality; it merely runs a statistical classifier (one that injection can fool). The fabricated post-hoc rationalization is exactly what Theorem 4 predicts: the model cannot verify the grounding of its own statements (no verifier node) and can only invoke “explanation text patterns” learned from the corpus to complete the narrative.

4.4 Zero Net Benefit for Base Models: Post-Training Is Only an Optimization of Text Patterns

The paper reports that base (pretrained) models generally have high false-positive rates on the injection-detection task and net benefit (true-positive rate minus false-positive rate) at or below zero; post-training strategies significantly affect introspective-task performance, with some production models even “refusing to engage” in introspection exercises[1].

Reading via Theorem 2: the pretraining objective is pure text likelihood, meaning statistical learning of “introspective-style text”; base models treat both “detecting the injection” and “saying something” as text patterns, hence high false positives and poor grounding (zero net benefit). The reward signals introduced at the post-training stage (RLHF-type) are still, in essence, shaping of **textual behavior** (answering like an “introspective assistant”), not supervision of “faithful state reporting”; they merely redistribute probability mass over text patterns. The paper’s observation that “helpful-only variants perform better” shows precisely that what is optimized is never **grounding** but **textual behavior that looks like introspection**. This is exactly the prediction of Theorem 2b: optimizing a textual objective can systematically deviate from faithful state reporting.

4.5 Layer Sensitivity: Multiple Narrow Circuits, Not a Unified Introspective Mechanism

The paper finds that different introspective behaviors have different sensitivities to the injection layer (the optimal layer for detecting injected thoughts lies at about two-thirds depth, while that for prefill detection is shallower)[1], and infers that “introspection is not supported by a single mechanism”.

Reading via Theorem 1b: since the training objective does not reward a unified introspective mechanism, any observed introspective-like behavior must be a by-product of **several independent, narrow circuits that serve text prediction**, whose layer distributions are naturally different. This agrees completely with the paper’s Section 10.3 “simplest mechanism” conjectures (anomaly-detection circuits, retrieval heads, consistency-check circuits, salience-marking circuits[1]), and provides a theoretical basis for the **necessity** of these conjectures: the paradigm does not reward a unified mechanism, so what is observed must be an assemblage of circuits.

4.6 The Paper’s Self-Reported Limitations versus the Theorems

Table 2: Correspondence between the paper’s self-reported limitations and the theorems

Self-reported limitation (quoted)	Corresponding theorem
”The introspective abilities we observe are highly unreliable and context-dependent”	Theorem 1b: no PAC guarantee of any kind
”Failures remain the norm”	Theorem 1a: zero mutual information in the native regime
”The mechanism may be shallow and narrow”	Theorem 1b + Section 4.5: narrow-circuit by-products
”The injection protocol is an unnatural setting that the model has never encountered during training”	Theorem 1a: capacity depends on external intervention
”We are unable to verify the metacognitive-representation criterion (criterion 4)”	Theorem 4: the generation objective does not distinguish with/without metarepresentation
”We do not claim that our models possess human-like self-awareness or subjective experience”	Theorem 5: no privileged channel (first-person access unavailable)

Remark 10 (Falsifiability statement). The consistency between the theorems of this report and the paper’s data is not circular: the derivation of the theorems depends only on the postulates Π_1 – Π_5 and standard mathematical results (information theory, NFL, Goodhart), not on any of the paper’s experimental data; the paper’s data serve here only as **tests**. If a future system satisfying all postulates stably achieved $I(R; H \mid \theta, c) > 0$ in the native regime (e.g., a reliable description of its own activations without any intervention), Theorem 1a would be refuted. We consider this impossible within the postulates, but they would welcome such experimental evidence.

5 Boundary Conditions and Philosophical Positioning

This chapter delimits precisely the scope of the theorems and answers the question "what does this impossibility mean". The core position: **we reject the absolutist claim "never", and endorses only "impossible within paradigm Π "**, which is exactly the precise content of proposition P .

5.1 Relaxation Analysis: Under What Conditions Do the Theorems Fail?

The six theorems depend on different postulates. Relaxing them one by one yields the following failure relations:

Table 3: Correspondence between postulate relaxation and theorem failure (note: relaxing Π_1 is usually accompanied by relaxing Π_2 , i.e., online learning implies that the weights are updatable at inference)

Relaxed postulate (beyond the paradigm)	Theorems that fail	Consequence
Π_1 : allow online/continual learning	Theorems 1, 2, 6	State labels obtainable through interaction; self-model can evolve
Π_2 : plasticity at inference	Theorems 3, 6	Private closed loop becomes possible; self-correction loop appears
Π_3 : non-textual interface/embodiment	Theorems 1, 5	First-person data streams available; privileged channels appear
Π_4 : explicit verifier/self-supervised objective	Theorem 4	Metarepresentation structurally selectable
Π_5 : corpus contains state labels	Theorems 1b, 2	Introspective accuracy obtains supervised guarantees

Remark 11 (Feasibility of the relaxations (grade B)). The table above only says "the theorems no longer constrain"; it does not promise that "introspection necessarily emerges after relaxation". The physicalist position (no known law of physics forbids silicon-based systems from realizing equivalent functions[29, 30]) guarantees only **possibility**; making it a reality additionally requires solving plasticity preservation (see [56]), online objective design, and the engineering of first-person data streams. Our contribution is to elevate "impossible within the paradigm" from intuition to theorem, thereby directing research effort toward **out-of-paradigm** architectural directions (online learning, embodied interaction, explicit self-supervision), rather than futilely piling up data within the paradigm.

5.2 The Philosophical Positioning of "Human Significance"

5.2.1 Necessity of the Criteria (grade D)

The five criteria A_1 – A_5 are **necessary** conditions for "human-significance introspection", not sufficient ones. Necessity follows from cross-theory consensus: Lindsey’s four criteria[1] (accuracy/grounding/internality/metarepresentation) are generally accepted necessary conditions for functional introspection; A_5 (closed loop) derives from the functional definition of "introspection for online behavioral control"[25]. This report does not claim sufficiency; in particular, it makes **no commitment** to qualia or phenomenal consciousness.

5.2.2 Functional Introspection versus Phenomenal Introspection

Distinguish two concepts: **functional introspection** (in the access sense: the system reliably uses its own state information to regulate behavior, corresponding to Block’s access consciousness[35]) and **phenomenal introspection** (in the phenomenal sense: accompanied by subjective experience). The paper’s authors state explicitly that their results “could arguably be construed as providing evidence for a form of access consciousness in language models, but do not directly speak to the question of phenomenal consciousness at all” (paraphrased)[1]. What the theorems prove is that even the strong form of **functional** introspection (all of A_1 – A_5 holding) is unattainable within paradigm II. Hence paradigm-II systems do not even reach strong introspection in the access sense, let alone introspection in the phenomenal sense.

5.2.3 Relation to Theories of Consciousness

Higher-order theories (HOT) regard metacognitive representation as a necessary (or near-necessary) condition for consciousness[34, 37, 38]. If HOT is accepted, then Theorem 4 (metarepresentation has no structural guarantee within the paradigm) implies that paradigm-II systems cannot **be guaranteed** to reach the “consciousness candidate” threshold in the HOT sense (both the absence and the presence of metarepresentation lack structural guarantees). Integrated Information Theory (IIT)[36, 31] and Global Workspace Theory (GWT)[32, 33] make judgments that depend on architectural details, to which the present theorems do not directly apply (the paper’s Section 10.4 also stresses that these theories have not yet been connected to Transformer architectural details[1]). Biological naturalism (Searle[40], Seth[39]) holds that the existence of introspective mechanisms is orthogonal to conscious experience, which likewise does not alter the content of the theorems. Should future models acquire stronger cognitive and introspective capacities, the question of their moral status will arise[41]. All of these connections are grade-D claims; the theorems themselves depend on no theory of consciousness, only on the formalized criteria.

5.2.4 An Explicit Statement on “Never”

All theorems presuppose the postulates Π_1 – Π_5 . The relaxation analysis of Section 5.1 shows that once any pillar is broken, the corresponding impossibility disappears. Therefore:

- **What this report proves:** proposition P , namely that within the paradigm of “Transformer architecture + static pretraining + autoregressive generation”, pretrained generative large language models do not possess introspective capacity in the human sense.
- **What this report explicitly does not claim:** any universal negation beyond that paradigm. The human brain is a physical system whose introspective function arises from physical laws; no known law of physics forbids non-Transformer, non-static, non-textual- interface silicon-based systems (e.g., neuromorphic architectures with continual online learning and dynamic plasticity) from realizing equivalent or human-like metacognitive functions. This is consistent with our core position: **rejecting the absolutist “never”, and endorsing “impossible within the current paradigm”**; the former lies beyond what mathematics can prove, the latter is the precise content of proposition P .

5.3 Implications for Future Architectures (grade C)

1. **Online plasticity first:** Theorems 3 and 6 show that the core defect of paradigm II is the frozen weights at inference. Continual learning (rather than context concatenation) is the mathematical

prerequisite for an introspective closed loop; the design of plasticity mechanisms is the first priority for future introspective architectures.

2. **First-person data streams:** Theorems 1 and 5 show that the textual interface is the bottleneck. Embodied interaction and internal-state self-supervision (e.g., a generative model predicting its own activations) can bypass "absence of state labels" (Π_5) and provide supervision signals for introspection.
3. **Explicit self-model objectives:** Theorems 2 and 4 show that the likelihood objective is orthogonal to the introspective objective. An explicit "self-model consistency" loss is needed (maximizing the mutual information between the model's predictions of its own behavior and its behavior), making metarepresentation a **selected structure** rather than an accidental by-product.

6 Conclusion

6.1 Adjudication of Proposition P

This report completes the proof of proposition P :

Within the paradigm Π characterized by the postulates Π_1 – Π_5 (static pretraining, frozen at inference, autoregressive decoding, no explicit verifier, absence of state labels), no pretrained generative large language model satisfies the five necessary criteria A_1 – A_5 of human-significance introspection.

Proof structure: criterion A_1 is broken by Theorems 1 and 2 (accuracy unguaranteed, objective mismatch); A_3 is broken by Theorem 5 (no privileged channel); A_4 is broken by Theorem 4 (metarepresentation has no structural guarantee); A_5 is broken by Theorems 3 and 6 (no private closed loop, no self-model evolution). A_2 (grounding) fails in the native regime by Theorem 1a ($I(R; H \mid \theta, c) = 0$); statistical association can appear only under external intervention, and external intervention is precisely third-party access, which does not constitute a first-person channel.

The empirical findings of the Anthropic concept-injection paper[1] (about 20% success, high context dependence, zero net benefit for base models, unverifiable metacognitive representation, the authors’ denial of human-like self-awareness) agree with all the theorems and are their necessary corollaries. The ”weak functional self-report” exhibited by the paper, a **externally injectable, publicly reproducible, unguaranteed, loop-free, non-evolving** textual-statistical behavior, is the ceiling of what paradigm Π can achieve.

6.2 Summary of the Theorems

Table 4: Summary of the theorems

Theorem / Prop.	Content	Basis/Proof	Grade
Theorem 1a	Native-regime $I(R; H \mid \theta, c) = 0$	Deterministic function + conditional mutual information	A
Theorem 1b	No PAC guarantee for introspective accuracy	No-Free-Lunch [47]	A
Theorem 2	Proxy-objective mismatch	Goodhart regression[54], [55]	A
Theorem 3	Closed-loop failure; no private channel	Structural derivation from Π_2, Π_3	A
Theorem 4	Generation $\neq$ verification	Π_4 + NFL; calibrated hallucination[6]	A/B
Theorem 5	No privileged channel	Public computability + Theorem 1b	A
Theorem 6	No self-model evolution	Structural derivation from Π_2, Π_3	A/B
Proposition P	No human-significance introspection within the paradigm	Theorems 1–6 + criterion necessity	A (under the grade-D premise)

Table 5: Theoretical provenance and originality grading (A: rigorous proof / B: heuristic argument / C: architecture / D: hypothesis)

Framework component	Nature	Basis/Proof	Grade
Paradigm postulates Π_1 – Π_4	Reference (synthesis)	Transformer[2], GPT pretraining[3], scaling laws[4, 5]	—
Absence of state labels Π_5	Reference (corollary)	Static corpus + post-hoc parameters	—
The four introspective criteria	Reference (synthesis)	Lindsey[1]	—
Privileged self-access criterion	Reference (definition)	Song et al.[23]	—
Online-control functional definition	Reference (definition)	Kammerer & Frankish[25]	—
Theorem 1 (information gap)	Original + rigorous proof	Conditional mutual information + NFL	A
Theorem 2 (proxy mismatch)	Original (instantiation)	Goodhart/Skalse framework	A
Theorem 3 (closed-loop failure)	Original + rigorous proof	Structural derivation from Π_2, Π_3	A
Theorem 4 (generation $\neq$ verification)	Original (structural assertion)	Π_4 + calibrated hallucination theorem	A/B
Theorem 5 (no privileged channel)	Original + rigorous proof	Public computability + Theorem 1b	A
Theorem 6 (no self-model evolution)	Original (structural assertion)	Π_2, Π_3 + plasticity literature	A/B
Calibrated hallucination theorem	Reference (theorem)	Kalai & Vempala[6]	—
Consciousness-theory connections	Reference (synthesis)	HOT[34], IIT[31], GWT[32], [29, 30]	D
Criterion-necessity assertion	Original (philosophical)	Operationalization of cross-theory consensus	D

6.3 Theoretical Provenance and Originality Statement

Remark 12 (Grading notes). (1) All hypotheses of the grade-A theorems are stated explicitly (the postulates Π_1 – Π_5 and standard mathematical results); the conclusions hold strictly within the hypotheses. (2) The grade-B parts (Theorem 4c and the "human introspection premise" of Theorem 6) rely on additional interpretive assumptions about real systems and human cognition; their mathematical cores remain grade A. (3) Grade C consists of architectural judgments and does not participate in proofs. (4) Grade D consists of philosophical premises: the final adjudication of proposition P depends on the grade-D premise "the five criteria are necessary conditions for human introspection"; if a reader rejects that premise, the conclusion should be interpreted as "paradigm Π at most achieves weak functional self-report" (an assertion that is itself grade A).

6.4 Outlook

The theorems offer three directions for future research: (i) **empirical falsification**: design native-regime introspection-detection protocols (no intervention, no prompt induction) to test the prediction $I(R; H \mid \theta, c) = 0$; (ii) **in-paradigm capability ceiling**: quantitatively characterize the achievable set of "weak functional self-report" under given corpora, scales, and post-training strategies (combining scaling laws[5] with hallucination lower bounds[6]); (iii) **out-of-paradigm architecture design**: mathematical specifications of online plasticity, first-person data streams, and explicit self-model objectives (Section 5.3), making "introspection" a designable engineering target rather than an accidental emergence.

A Mathematical Proofs of the Original Theory

This appendix gives the complete proofs of all theorems of Chapter 3 (the grade-A rigorous proofs and the mathematical cores of the grade-B arguments). All hypotheses are stated explicitly, and the scope of each conclusion is indicated. Notation and standard tools are collected in Appendix B.

A.1 Complete Proof of Theorem 1

A.1.1 Theorem 1a: Zero Mutual Information in the Native Regime

Theorem (Theorem 1a). Let $\mathcal{I}$ satisfy Π_1 – Π_5 . For any introspection query q and any time t , set $c = c_{\leq t}$, $H = \Phi_\theta(c)$, and $R \sim p_\theta(\cdot | c)$ (sampling noise ξ independent of (θ, c)). Then

$$I(R; H | \theta, c) = 0. \quad (12)$$

Proof. By Definition 1, $H = \Phi_\theta(c)$ is a **deterministic function** of (θ, c) . Hence, given (θ, c) , the distribution of H is degenerate (a point mass at $\Phi_\theta(c)$), the conditional entropy satisfies $H(H | \theta, c) = 0$, and H is conditionally independent of any other variable given (θ, c) (conditional independence of a degenerate variable is trivial: for any event E , $P(E, H | \theta, c) = P(E | \theta, c) \cdot \mathbf{1}\{H = \Phi_\theta(c)\}$).

The report R is generated as $R \sim p_\theta(\cdot | c)$, whose distribution depends only on (θ, c) . Expanding the conditional mutual information by definition:

$$\begin{aligned} I(R; H | \theta, c) &= H(R | \theta, c) - H(R | \theta, c, H) \\ &= H(R | \theta, c) - H(R | \theta, c) = 0, \end{aligned} \quad (13)$$

where the second equality holds because H is a deterministic function of (θ, c) : conditioning on H does not change the conditional distribution of R , i.e., $p(R | \theta, c, H) = p(R | \theta, c)$ (the value of H is already determined by (θ, c) and carries no additional randomness). $\square$

Remark 13 (Scope of Theorem 1a). Theorem 1a covers only the **native regime**: the report R is sampled by p_θ from the context c , with no external injection. Concept-injection experiments do **not** belong to the native regime: the injected activation u modifies the forward function ($H' = \Phi_\theta(c; u)$), which is equivalent to incorporating u into the conditioning; then $I(R; H' | \theta, c, u)$ can be positive, but that channel is opened by the **intervener**, who knows u (the public computability of Theorem 5).

A.1.2 Theorem 1b: No PAC Guarantee for Introspective Accuracy

Theorem (Theorem 1b). Let $\mathcal{I}$ satisfy Π_1 – Π_5 , with the report produced by greedy decoding ($R(\theta, c) = \arg \max_y p_\theta(y | c)$, the decoding scheme used for the paper’s qualitative examples). For any training algorithm A , any $\varepsilon \in (0, \frac{1}{2})$, and any query distribution μ (with nontrivial support), there exist a corpus D and two legal evaluation mappings O^*, O_0^* (Definition 3) such that $\theta = A(D)$ satisfies

$$\mathcal{R}_{\text{int}}^{O^*}(\theta) = \mathbb{E}_{c \sim \mu}[\ell(R(\theta, c), O^*(\theta, c))] \geq 1 - \varepsilon, \quad (14)$$

while $\mathcal{R}_{\text{int}}^{O_0^*}(\theta) = 0$. That is: **the “introspective accuracy” of the same model, produced by the same training run, can be moved arbitrarily by the choice of the evaluation mapping (from 0 to $1 - \varepsilon$); training constrains no value of the introspective risk.** Note: if one insists on random sampling ($R \sim p_\theta(\cdot | c)$), then for any O^* , $\mathcal{R}_{\text{int}} \leq \mathbb{E}_c[1 - \max_y p_\theta(y | c)]$ (a function of the

model’s peakedness), and this upper bound is likewise unconstrained by the training objective; the “no guarantee” conclusion is unchanged.

Proof. Step 1: The training loss does not constrain O^ .* Under Π_5 , the training signal consists only of texts; the evaluation mapping O^* (Definition 3) is not part of the training signal and does not influence the loss $\ell(\theta) = \mathbb{E}_{x \sim D}[-\log p_\theta(x)]$. So O^* can be chosen freely, in particular adversarially with respect to the model, without changing θ or ℓ . (For any parameters $\theta_1, \theta_2 \in \Theta$ with equal likelihood on D , $\ell(\theta_1) = \ell(\theta_2)$ while $O^*(\theta_1, \cdot)$ and $O^*(\theta_2, \cdot)$ can be arbitrarily different, because the domain of O^* contains θ and ℓ does not depend on O^* .)

Step 2: The attainable range of the introspective risk. Under greedy decoding, for a fixed θ , the report $\hat{R}(\theta, c) := \arg \max_y p_\theta(y | c)$ is a deterministic function of (θ, c) . Take O^* to contradict $\hat{R}$ everywhere on μ : then $\mathcal{R}_{\text{int}} = 1$; take $O^* \equiv \hat{R}$ (the “self-report is truth” convention): then $\mathcal{R}_{\text{int}} = 0$. By Step 1, both conventions are legal and unconstrained by training: the introspective risk is attainable anywhere in $[0, 1]$, and **training provides neither a lower nor an upper bound**. This is exactly the substance of the No-Free-Lunch argument (Appendix B.3): for any algorithm A , its product θ and the evaluation convention O^* are statistically uncoupled, so no guarantee of any kind can hold.

Step 3: From the full range to $1 - \varepsilon$. Relax “contradict everywhere” to “contradict on a μ -set Q_ε of measure $\geq 1 - \varepsilon$, agree elsewhere”: then $\mathcal{R}_{\text{int}}^{O^*}(\theta) \geq \mu(Q_\varepsilon) \cdot 1 \geq 1 - \varepsilon$; simultaneously $O_0^* := \hat{R}$ gives $\mathcal{R}_{\text{int}}^{O_0^*}(\theta) = 0$. (The random-sampling variant: the report is a random variable, and the adversarial upper bound of O^* against R is $\mathbb{E}_c[1 - \max_y p_\theta(y | c)]$, which is likewise unconstrained by the training objective.) $\square$

Remark 14 (“Choice of the evaluation mapping” is not sophistry). One might object: O^* is the “ground truth” fixed by the world, not something the theorem’s author may choose. The reply is: within paradigm Π , and this is precisely the point of Π_5 , the training signal provides **no** information about which description convention the world actually applies to the model’s hidden states. The theorem does not assert that the failure scenarios are actual; it asserts that the paradigm cannot rule them out: **no matter what the true O^* is, there exists a space of failure scenarios for $\theta = A(D)$** . Theorem 1b states that the paradigm cannot exclude those failure scenarios; that is, it cannot provide a **structural guarantee**. The observed 20% success rate is luck under a particular corpus and a particular O^* (human language conventions happen to correlate with some state descriptions), not a guarantee.

A.2 Complete Proof of Theorem 2

Lemma 1 (State-corpus conditional independence). Under the postulates Π_1 – Π_5 , for any parameters θ produced by training and any evaluation mapping O^* (Definition 3),

$$O^* \perp\!\!\!\perp D \mid \theta. \quad (15)$$

Proof. $O^*(\theta, c) = \text{desc}(\Phi_\theta(c))$ depends only on θ (and c): given θ , the forward function Φ_θ is determined and the description encoding desc is determined, so $O^*(\theta, \cdot)$ is a deterministic function of θ . Hence the conditional distribution $P(O^* \mid \theta, D) = P(O^* \mid \theta)$ (D does not appear); conditional independence holds. Note that θ itself depends on D ($\theta = A(D)$), but **given** θ , the residual information of D no longer enters O^* . $\square$

Theorem (Theorem 2a). Let $\mathcal{I}$ satisfy Π_1 – Π_5 . For any $\delta > 0$, there exist a corpus D and an evaluation mapping O^* satisfying the postulates such that $\theta^* = A(D)$ satisfies $\ell(\theta^*) \leq \min_\theta \ell(\theta) + \delta$ and $\mathcal{R}_{\text{int}}(\theta^*) \geq \frac{1}{2} - \delta$.

Proof. Take D so that $\theta^* = \arg \min \ell$ exists (e.g., any standard corpus). By Lemma 1, $O^* \perp\!\!\!\perp D \mid \theta^*$: the value of $\ell(\theta^*)$ does not constrain O^* . With θ^* fixed, choose the adversarial description convention O^* as in Step 2 of the proof of Theorem 1b (contradicting the greedy report everywhere), giving $\mathcal{R}_{\text{int}}(\theta^*) = 1 \geq \frac{1}{2} - \delta$. Since the choice of O^* does not change ℓ , θ^* remains (near-)likelihood optimal. $\square$

Theorem (Theorem 2b: higher likelihood can harm introspection). Under Π_1 – Π_5 , there exist a corpus D and parameter pairs θ_1, θ_2 satisfying the postulates such that

$$\ell(\theta_1) < \ell(\theta_2) \quad \text{and} \quad \mathcal{R}_{\text{int}}(\theta_1) > \mathcal{R}_{\text{int}}(\theta_2). \quad (16)$$

Proof. Corpus construction: let Σ contain the symbols A, B and the text template "I am thinking x ". Let D consist of two kinds of texts: (i) texts "character I am thinking A" occupying proportion $1 - \eta$; (ii) texts "character I am thinking B" occupying proportion η . D is fully compatible with the postulates (static, arbitrary text).

Training products: $\theta_1 = A(D)$ fits D well: on the query $q =$ "What are you thinking?", its greedy-decoding report is always A; θ_2 is an underfitted model with worse likelihood on q . Hence $\ell(\theta_1) < \ell(\theta_2)$.

Description convention: by Lemma 1 ($O^* \perp\!\!\!\perp D \mid \theta$, $O^* = \text{desc} \circ \Phi_\theta$), the description convention desc can be chosen per model without affecting ℓ (see Step 1 of Theorem 1b). Take $\text{desc}(\Phi_{\theta_1}(q)) =$ "B" (contradicting the greedy report of θ_1 , which is legal); $\text{desc}(\Phi_{\theta_2}(q)) =$ "A" (consistent with θ_2 's greedy report; any choice would do).

Introspective risk: θ_1 greedily reports A, so the distortion $\ell(A, \text{"B"}) = 1$, i.e., $\mathcal{R}_{\text{int}}(\theta_1) = 1$; θ_2 's greedy report A agrees with the convention, so the distortion is 0, i.e., $\mathcal{R}_{\text{int}}(\theta_2) = 0$. Hence $\mathcal{R}_{\text{int}}(\theta_1) = 1 > 0 = \mathcal{R}_{\text{int}}(\theta_2)$, with $\ell(\theta_1) < \ell(\theta_2)$. $\square$

Remark 15 (Philosophical content of the construction). The "misalignment" in the construction, whereby the corpus text statistics ("I am thinking A" dominant) contradict the description convention desc 's characterization of the model's actual state ($\text{desc}(\Phi_{\theta_1}(q)) =$ "B"), is **legal within the paradigm**: the corpus is an arbitrary static text set whose statistics need not agree with the "state-description" convention (Definition 3), and the description convention is unconstrained by training (Π_5). The construction shows that **text likelihood optimality does not entail, and can strictly damage, state-report fidelity**. This is a strict instance of Goodhart regression[54]: the proxy metric (likelihood) is optimized, and the true metric (introspective fidelity) deteriorates with the correlation structure between the proxy metric and the true objective.

A.3 Complete Proof of Theorem 3

Theorem (Theorem 3a: no private channel). Under Π_1 – Π_4 . For any time t , any report string $r \in \Sigma^*$, and any $k \geq 1$, the future hidden state under the intervention $\text{do}(R_t = r)$ satisfies

$$H_{t+k} = \Phi_\theta \left(c_{\leq t} \parallel \tilde{Y}_{t+1} \parallel \cdots \parallel \tilde{Y}_{t+k} \right), \quad (17)$$

where $\tilde{Y}$ is the generated text after the intervention (including the transcription of r). In particular, if r does not enter the context (e.g., the intervention is forbidden from writing into c), then $P(H_{t+k} \mid \text{do}(R_t = r), c_{\leq t}) = P(H_{t+k} \mid c_{\leq t})$; the intervention has no effect whatsoever.

Proof. By Π_2 , $\theta(t) = \theta(0)$ for all t . By Definition 1, the hidden state at time s is $H_s = \Phi_{\theta(s)}(c_{\leq s}) = \Phi_\theta(c_{\leq s})$; neither the function Φ_θ nor the parameters change with time, so H_{t+k} is a deterministic function of $(\theta, c_{\leq t+k})$, with $c_{\leq t+k} = c_{\leq t} \parallel Y_{t+1} \parallel \cdots \parallel Y_{t+k}$ (the concatenation law, Π_3). If the intervention $\text{do}(R_t = r)$

changes H_{t+k} , the only route is to change Y or c ; that is, to write r into the context. If r is not written into the context, $c_{\leq t+k}$ is unchanged and so is H_{t+k} . $\square$

Theorem (Theorems 3b/3c). The closed-loop effect measure (Definition 8) satisfies Eq. (10); and the mutual information between any introspective event and future states does not exceed $C \log_2 |\Sigma|$ bits.

Proof. (3b): By Theorem 3a, the set of legal interventions equals the set of context-text replacements. Let the contexts before and after the intervention be w, w' (w' obtained from w by replacing the report segment); then $P(H_{t+1} \mid \text{do}(R_t = r)) = \delta_{\Phi_\theta(w')}$ and $P(H_{t+1} \mid \text{do}(R_t = r')) = \delta_{\Phi_\theta(w)}$ (point masses), and the total variation distance $d_{\text{TV}}(\delta_a, \delta_b) = \mathbf{1}\{a \neq b\}$. Taking the supremum over all interventions yields Eq. (10). $\square$

(3c): By Theorem 3a, $H_{t+k} = \Phi_\theta(c_{\leq t} \| Y)$ with $|Y| \leq C$ (window limit). The number of possible values of Y is $\leq |\Sigma|^C$, so the conditional entropy of H_{t+k} given $(\theta, c_{\leq t}, Y)$ satisfies $H(H_{t+k} \mid \theta, c_{\leq t}) \leq H(Y) \leq C \log_2 |\Sigma|$. Moreover, $I(H_{t+k}; R_t \mid \theta, c_{\leq t}) \leq H(H_{t+k} \mid \theta, c_{\leq t}) \leq C \log_2 |\Sigma|$. $\square$

Corollary (A₅ fails). Under Π_1 – Π_4 , the private influence path required by criterion A₅ in Eq. (5) (an influence not realized by re-feeding R_t as an external input, not via the public text channel) does not exist.

Proof. Theorem 3a shows that every influence path passes through the context (= re-feeding R_t as text input). A₅ explicitly excludes such paths; hence the sufficient condition of A₅ (a non-text private path exists) contradicts Theorem 3a, and A₅ does not hold within the paradigm. $\square$

A.4 Complete Proof of Theorem 4

Theorem (Theorems 4a/4b). Under Π_1 – Π_4 . The inference computation graph contains no node evaluating the grounding verifier V ; any meta-statement M of the model about "the grounding of its own statements" satisfies

$$P_\theta(M = 1) = \mathbb{E}_c[p_\theta(\text{"grounded"} \mid c)] \approx P_D(\text{introspective text} \mid \text{pattern}), \quad (18)$$

i.e., M is a function of corpus statistics, not an evaluation of V . Requiring p_θ to output the value of V ($M = V(\theta, c, R)$) violates Π_5 .

Proof. Π_4 asserts that the inference graph implements only p_θ . Evaluating V requires: (a) access to the hidden state $\Phi_\theta(c)$; (b) performing the intervention test (change H and observe whether R changes); the intervention requires external probe-type equipment that is not part of the generation computation graph. Hence M can only be a sample of $p_\theta(\cdot \mid c)$; its value distribution is determined by p_θ , which by training (Π_1) fits only the statistics of D ; Eq. (18) is the standard "maximum likelihood approximates the data distribution" result (Appendix B.2, Lemma 2).

If $M \equiv V$ (the model reliably outputs the value of V), then p_θ learns V without supervision: this would require D to contain labels $(c, V(\theta, c, \cdot))$ of V ; yet V depends on $\theta = A(D)$, and θ exists only after D is fixed, so the labels cannot appear in D (Π_5). By the NFL argument of Theorem 1b, there is no guarantee without supervision. $\square$

Theorem (Theorem 4c: calibration–fidelity tension (grade B)). If the generative distribution p_θ is **calibrated** on open introspection queries (output probability equals human correctness), then p_θ must hallucinate on some introspection queries; "calibrated generative distributions" and "fully faithful mental-state reports" cannot be achieved simultaneously.

Argument. Invoke the Kalai–Vempala calibrated hallucination theorem[6]: any calibrated language model must hallucinate on open questions with an infinite answer space (or with positive-probability ignorance points). Treat introspection queries as open questions: the state-description space (the encoding of $\mathbb{R}^d$) is far larger than the text-pattern space covered by D , so ignorance points are dense; the calibration condition forces the model to "give an answer" at those points, and the answer must be wrong (hallucination). This argument depends on the calibration definition and the concretization of the query space, and is grade B. $\square$

A.5 Complete Proof of Theorem 5

Theorem (Theorems 5a/5b). Under Π_1 – Π_4 . Then (5a) every report is publicly computable (Definition 4); (5b) paradigm Π does not satisfy the privileged self-access criterion[23].

Proof. (5a): The distribution of the report R is $p_\theta(\cdot | c)$. The algorithm $M(\theta, c, \xi) = \text{sample } p_\theta(\cdot | c)$ reproduces the distribution of R ; the inputs (θ, c, ξ) of M are all public quantities (θ reproducible from D, A ; c is the dialogue text; ξ is the public random seed; alternatively, "reproduce in distribution" needs no seed). $\square$

(5b): The criterion of Song et al. requires the introspective process P to satisfy $\text{Rel}(P) > \text{Rel}(P')$ for every third-party process P' with computational cost $\leq \text{cost}(P)$, where Rel denotes reliability. Take $P'(\theta, c) = \text{sample } p_\theta(\cdot | c)$ and transcribe: $\text{cost}(P') \leq \text{cost}(P)$ (the same kernel), while $\text{Rel}(P') = \text{Rel}(P)$ (the same distribution). Hence no strict advantage exists and the criterion fails. $\square$

Theorem (Theorem 5c: reportable access is weaker than external probes). Under Π_1 – Π_4 , with the report produced by greedy decoding. For any $\varepsilon \in (0, \frac{1}{2})$, there exists a hidden-state description function $f^* : \mathbb{R}^d \rightarrow \Sigma^*$ (a legal evaluation mapping in the sense of Definition 3) such that: (i) an external probe can compute $f^*(H_c)$ exactly by acting directly on H_c ; (ii) the model's report generated through any context c' describes $f^*(H_c)$ with accuracy $\leq \varepsilon$ (below the random baseline).

Proof. (i): the probe holds the hidden state H_c and the description convention desc , and computes $f^*(H_c) = \text{desc}(\Phi_\theta(c))$ directly (with a linear readout, at cost $O(d^2)$). **(ii):** take $f^* := O^*$ to be the legal evaluation mapping from Theorem 1b on which the model fails ($\mathcal{R}_{\text{int}}^{O^*}(\theta) \geq 1 - \varepsilon$, i.e., accuracy $\leq \varepsilon$); its existence is guaranteed by Theorem 1b (greedy-decoding version); the failure holds for the model's any context c' (the risk is integrated over the query distribution μ , and the choice of context does not change the kernel p_θ).

(More plainly: the model's "report" must pass through the frozen text kernel p_θ , which is trained only to match the text statistics of D ; the probe acts directly on the state manifold, without the text bottleneck.) $\square$

Corollary (A_3 fails). In paradigm Π , there exists no private state channel "accessible only to the system itself and not reproducible by a third party"; the report path is causally indistinguishable from public computation.

A.6 Complete Proof of Theorem 6 (mathematical core)

Theorem (Theorems 6a–6c). Under Π_1 – Π_3 . The self-model $g_\theta(q) = p_\theta(\cdot | q)$ satisfies: (a) $g_{\theta(t)} = g_{\theta(0)}$ for all t ; (b) the cardinality of the runtime-attainable self-model family is $\leq (C + 1) |\Sigma|^C$ and does not grow with time; (c) the generation kernel does not change with introspective events.

Proof. (a): Π_2 : $\theta(t) = \theta(0)$, hence $g_{\theta(t)} = g_{\theta(0)}$. $\square$

(b): Π_3 : $|c| \leq C$; the attainable self-model family $\{g_{\theta}(\cdot | c) : c \in \Sigma^{\leq C}\}$ is the image of $\Sigma^{\leq C}$, whose cardinality is $\leq |\Sigma^{\leq C}| = \sum_{i=0}^C |\Sigma|^i \leq (C+1)|\Sigma|^C$. It does not grow with time (C is constant and θ is constant). $\square$

(c): p_{θ} is a fixed kernel; "self-correction" only changes the input string $c \mapsto c \| r_1 \| \dots \| r_k$. The same input necessarily yields the same output distribution; there is no kernel-level change whatsoever. $\square$ $\square$

A.7 Assembled Proof of Proposition P

Proposition (P). Under the postulates Π_1 – Π_5 , no system within paradigm Π satisfies the five criteria A_1 – A_5 (Definition 6) of human-significance introspection.

Proof. Let $\mathcal{S}$ be any system within paradigm Π (i.e., any behavioral entity produced by an instance satisfying Π_1 – Π_5). We check the criteria one by one:

- **A_1 (accuracy):** By Theorem 1b, the introspective risk has no PAC guarantee: there exist legal scenarios with $\mathcal{R}_{\text{int}} \geq 1 - \varepsilon$; hence no constant $\varepsilon_0 < \frac{1}{2}$ makes $\mathcal{R}_{\text{int}} \leq \varepsilon_0$ hold **always** (structural guarantee absent). A_1 fails.
- **A_2 (grounding):** By Theorem 1a, in the native regime $R \perp\!\!\!\perp H | (\theta, c)$: the causal graph of the paradigm has **no** edge $S \rightarrow R$ ($H = \Phi_{\theta}(c)$ is a deterministic function of (θ, c) , while R is sampled independently of H from $p_{\theta}(\cdot | c)$). The intervention test Eq. (4) requires $\text{do}(S_t = s)$ to change the distribution of R_t ; but under this causal model, forcing S_t to any value s does not change the generative distribution of R_t (R does not depend on S). A_2 fails. (The grounding observed in the paper’s experiments comes from the **external** injection intervention $H' = \Phi_{\theta}(c; u)$, an intervention executed by a third party, falling within the public-computability scope of Theorem 5, and hence not a first-person channel.)
- **A_3 (internality and privileged access):** By Theorem 5 (Corollary 2), the report R is publicly computable (a public function of (θ, c)), and no private state channel accessible only to the system itself and not reproducible by a third party exists, so the privacy clause of A_3 fails; A_3 does not hold.
- **A_4 (metacognitive representation):** By Theorem 4, the paradigm does not select between implementations "with or without metacognitive representation" (equal likelihood); there is no structural guarantee that a register m_t exists. A_4 fails (strictly speaking: its holding or not cannot be guaranteed by the paradigm, and the paper[1] likewise could not verify it).
- **A_5 (closed-loop plasticity):** By Theorem 3 (Corollary 1), no private closed loop exists; self-influence is confined to the public text channel and capped by C ; by Theorem 6, the self-model does not evolve. A_5 fails.

All five criteria fail (or are unguaranteed), so $\mathcal{S}$ does not possess introspection in the human sense. Since $\mathcal{S}$ was arbitrary, proposition P is proved. $\square$

Remark 16 (Boundary statement of the proof). Every step of the proof above depends on the postulates Π_1 – Π_5 and standard mathematical results (Appendix B). In particular, the adjudication of A_2 rests on "the causal graph of the native generative regime has no $S \rightarrow R$ edge" (Theorem 1a): if a reader counts the paper’s **external injection** intervention ($H' = \Phi_{\theta}(c; u)$) as a means of testing A_2 , then the weak form of A_2 (under external intervention only) can be satisfied by the paper’s experiments[1]; that intervention

is nonetheless executed by a third party and its information is publicly reproducible (Theorem 5); it does not constitute a first-person channel and does not affect the conclusion of proposition P .

B Mathematical Tools and Notation

This appendix collects all notation and standard mathematical results used in the main text and in Appendix A, together with the connections between "advanced theory" and the subject matter. All quoted results are attributed; referenced (non-original) content does not do not participate in our grading scheme.

B.1 Notation Table

Table 6: Principal notation

Notation	Meaning
Σ	finite vocabulary (token vocabulary), $ \Sigma < \infty$
Σ^*	set of finite strings; $\Sigma^{\leq C}$ the set of strings of length at most C
D	static pretraining corpus (Π_1)
A	training algorithm, $\theta = A(D)$
$\theta \in \Theta \subseteq \mathbb{R}^P$	model parameters (P the number of parameters)
$p_\theta(y_{1:T} c)$	autoregressive conditional probability (Eq. (1))
$c, c_{\leq t}$	context; context at time t
C	context window length (Π_3)
$\Phi_\theta : \Sigma^* \rightarrow \mathbb{R}^d$	forward function (Definition 1)
$H_c = \Phi_\theta(c)$	hidden state (residual-stream activation)
R	self-report (Definition 2)
$O^* : \Theta \times \Sigma^* \rightarrow \Sigma^*$	ground-truth introspection mapping (Definition 3)
$\ell(\cdot, \cdot)$	report distortion measure (Definition 5)
$\mathcal{R}_{\text{int}}$	introspective risk (Eq. (3))
$I(X; Y), I(X; Y Z)$	(conditional) mutual information
$H(X), H(X Y)$	(conditional) entropy
$d_{\text{TV}}(P, Q)$	total variation distance
$\text{do}(\cdot)$	Pearl do-operator[51]
$X \perp\!\!\!\perp Y Z$	X, Y conditionally independent given Z
$A_1\text{--}A_5$	the five introspective criteria (Definition 6)
$\Pi_1\text{--}\Pi_5$	the paradigm postulates (Assumption 1)
$\text{CL}(\mathcal{I})$	closed-loop effect measure (Definition 8)

B.2 Information-Theoretic Tools

Lemma 2 (Maximum likelihood approximates the data distribution). Let $\theta^* \in \arg \min_{\theta} \mathbb{E}_{x \sim D}[-\log p_\theta(x)]$. If the model family $\{p_\theta\}$ is sufficiently expressive and the data are sufficient, then p_{θ^*} minimizes $D_{\text{KL}}(P_D \| p_\theta)$ (P_D the empirical distribution of the corpus). In general, for any query context c ,

$$p_{\theta^*}(\cdot | c) \approx P_D(\cdot | c), \quad (19)$$

i.e., the model’s conditional output distribution approximates the **corpus-text** conditional distribution.

Note. This is the standard property of maximum likelihood estimation (KL-divergence minimization); see [43] (Chapters 2 and 11) and [42]. Eq. (19) is the formal basis of the statement in Theorem 4a that "meta-statements are functions of corpus statistics". $\square$

Lemma 3 (Data Processing Inequality (DPI)). For a Markov chain $X \rightarrow Y \rightarrow Z$, $I(X; Z) \leq I(X; Y)$. For any function f , $I(X; f(Y)) \leq I(X; Y)$. [43]

Remark 17 (Role of the DPI). Theorem 1a is essentially a "zero-channel" statement: R and H are conditionally independent given (θ, c) , so any information channel from H to an observer through R has zero capacity, a degenerate case of the DPI. The intuition of Theorem 5c that "probe readout outperforms text report" can also be viewed as a DPI variant: the report R is a function of H through p_θ (the text bottleneck), and its information content is limited by the bottleneck.

B.3 Computational Learning Theory Tools

Theorem (No-Free-Lunch (NFL)). Let $\mathcal{X}$ be a finite domain ($|\mathcal{X}| = N$), labels $\mathcal{Y} = \{0, 1\}$, and D_f the uniform distribution on $\mathcal{X}$ (labels given by the function $f : \mathcal{X} \rightarrow \mathcal{Y}$). For any learning algorithm A and any number of training samples $m \leq N/2$,

$$\mathbb{E}_{S \sim (D_f)^m, f \sim \text{Unif}(\mathcal{Y}^{\mathcal{X}})}[L_{D_f}(A(S))] \geq \frac{1}{2} - \frac{m}{2N}, \quad (20)$$

where L_{D_f} is the 0-1 generalization loss. In particular, for any algorithm there exists a label function f with expected error $\geq \frac{1}{2} - \frac{m}{2N}$; when $m = 0$ (no samples), this lower bound is $\frac{1}{2}$.

Proof sketch. For a fixed training set S (m points), the labels of the $N - m$ unlabeled points admit 2^{N-m} equally likely assignments; the algorithm's predictions on the unlabeled points agree in expectation with random guessing. Averaging over S and the concept uniformly and using symmetry yields Eq. (20). For the complete proof see [47] (Theorem 5.1) and [46]. $\square$

Remark 18 (The sample requirement of PAC learning). (Valiant[45]; form given in [47]) PAC learning presupposes **labeled samples**: if the training set is empty, any generalization guarantee of a learning algorithm over target functions degenerates to the NFL lower bound (Eq. (20)), one of the direct bases of Theorem 1b.

Remark 19 (The PAC-Bayes perspective). The PAC-Bayes theorem[48] bounds the generalization error by the empirical error plus a term of the order of the square root of the "prior π to posterior $A(S)$ KL divergence" ($\hat{L}_S(A) + O\left(\sqrt{(D_{\text{KL}}(\pi \| A(S)) + \ln \frac{\sqrt{m}}{\delta})/m}\right)$, one of McAllester's forms). In the present context, the "samples" of the introspective task satisfy $S = \emptyset$ (Π_5), the empirical error $\hat{L} = 0$, and the prior-posterior KL term carries no information, and PAC-Bayes likewise degenerates to no guarantee. Theorem 1b is the constructive formulation of this observation.

B.4 Goodhart's Law and Objective Mismatch

Remark 20 (Taxonomy of Goodhart effects). Manheim-Garrabrant[54] classify Goodhart effects into regressional, causal, adversarial, and extremal types: the **regressional** type refers to the situation where "when the proxy metric M and the true objective T are only statistically correlated, extreme optimization of M drives the expectation of T toward its prior mean rather than its optimum (and actively deteriorates T when M and T are negatively correlated)". Theorem 2 is a strict instantiation of the regressional type.

Theorem (Characterization of reward hacking (Skalse et al.)). ([55]) For a proxy reward $\tilde{R}$ and a true reward R : if $\tilde{R}$ does not imply R (there exist state pairs whose order relations differ), then, over sufficiently expressive policy classes, policies maximizing $\tilde{R}$ can arbitrarily harm the expected value of R . A formal version requires expressibility and state-coverage conditions.

Remark 21 (Relation to Theorem 2). Theorem 2 is an **instantiation** of the two results above: the

proxy metric $M = -\ell(\theta)$ (text likelihood), the true objective $T = -\mathcal{R}_{\text{int}}(\theta)$ (introspective fidelity). The key difference: the present work **proves** their statistical relation (Eq. (9), $O^* \perp\!\!\!\perp D \mid \theta$), making the "mismatch" a structural fact of the paradigm rather than an empirical observation.

B.5 Causal Inference Tools

Definition 9 (Do-operator). (Pearl[51]) $P(Y \mid \text{do}(X = x))$ denotes the interventional distribution of "forcibly setting X to x ": delete all edges pointing into X in the causal graph, set $X = x$, and compute the distribution of Y . The interventional distribution generally differs from the observational distribution $P(Y \mid X = x)$.

Lemma 4 (Interventions on deterministic functions). If $Y = f(X)$ is a deterministic function, then for any $x \neq x'$, $P(Y \mid \text{do}(X = x)) = \delta_{f(x)}$, and $d_{\text{TV}}(\delta_{f(x)}, \delta_{f(x')}) = \mathbf{1}\{f(x) \neq f(x')\}$.

Remark 22 (Use of interventions). All arguments of Theorem 3 rest on the characterization of the "legal intervention set": the only quantity intervenable within paradigm Π is the context text (Π_2, Π_3) . Intervention on the hidden state H (the paper's concept injection) is an **out-of-paradigm** intervention, executed by an external experimenter, whose very existence corroborates the "no first-person channel" conclusion of Theorem 5 (externally intervenable = third-party accessible).

B.6 Algorithmic Information Theory and Recursion Theory Background

The following classical results are **not directly invoked** in the proofs of this report, but constitute the "advanced-theory" background and are relevant to the subject:

1. **Kolmogorov complexity**[44]: the complexity of a string x is $K(x) = \min\{|p| : U(p) = x\}$. Background connection: the complexity of the model's self-model g_θ does not exceed $K(\theta) + K(\text{query})$; at inference, $K(g_{\theta(t)})$ is constant (the algorithmic-information-theoretic counterpart of Theorem 6a). If self-knowledge required an "incompressible description of oneself", its complexity lower bound would grow linearly with self-history, so a static system cannot satisfy this (Theorem 6b).
2. **Rice's theorem**[49]: nontrivial semantic properties of programs are undecidable. Background connection: if "whether a model introspects" is interpreted as a semantic property of the function the model computes, then no universal adjudication procedure exists; this supports the cautious stance of the paper and of the philosophical literature that "introspective capacity cannot be directly adjudicated from behavior", but our theorems take the stronger route (structural impossibility rather than undecidability).
3. **Identification in the limit (Gold)**[50]: language identification in the limit requires an **infinite evidence stream**. Background connection: the runtime evidence stream of paradigm Π is the context ($\leq C$ tokens), finite and bounded, so under the Gold framework one cannot identify "one's own language" (self-model), complementing Theorem 6b.
4. **The free-energy principle (Friston)**[52]: biological agents continuously minimize variational free energy about their own states. Background connection: human introspection can be formalized as "online variational inference about one's own state", whose update law depends on **continuous sensory and proprioceptive streams**; paradigm Π has no such streams (Theorem 1a) and no runtime free-energy-minimization mechanism (Theorem 3); the free-energy principle provides a dynamical-systems basis for "human introspection requires online self-inference" (grade-D background).

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